Paper Review-Dixin Zhang

Title & Author

Agrawal, Anupam, Rajesh Mohanty, Shamik Bhattacharjee, and Abhimanyu Mittal. Hierarchical Contextual Uplift Bandits for Catalog Personalization. Preprint, arXiv:2601.14333v1, January 20, 2026.

Summary

This paper studies a personalization problem in fantasy sports where user behavior and platform conditions change very rapidly. To address that challenge, the authors propose a Hierarchical Contextual Uplift Bandit model that combines hierarchical context, uplift-based rewards and Bayesian UCB selection. The system doesn't simply order items; it alters parts of the contest catalog. In the paper, the authors report a 0.42% revenue gain in a 6 million user online A/B test, before deploying it to production where the revenue gain later grew to 0.51%.

Strengths

A key strength of the paper is that it solves a real production problem, instead of only being tested in simulation. The hierarchy is valuable as it will allow the model to use the general system level context when the user level evidence is weak but also allow for fine-grained personalization when the evidence is strong enough. The reward inheritance also sounds compelling: when the child node cannot produce a statistically significant uplift, it can inherit the reward from the parent node rather than make a decision with a few noisy pulls. This allows the approach to be usable in a cold-start setting, and in a changing production environment. The large-scale A/B test is also nice, as it provides more credibility to the business impact.

Major Comments

My biggest worry is about the reward function. The paper says that reward is a weighted sum of uplift to short term engagement, a proxy for long-term retention, and revenue, but I don't understand how they chose the weights and how sensitive the results are to the choice of weights. As the weights determine what the algorithm is optimizing for, this feels very underexplained.

Another issue is that the experimental section is promising but still somewhat limited. The online A/B test results are valuable, but the paper does not give much detail about segment-level effects, robustness over time, or whether the gains were consistent across different match contexts. Also, the offline simulation mainly compares HCUB with and without reward inheritance, but it would be stronger if the authors also separated the effects of the hierarchy itself, the uplift objective, and the Bayesian UCB decision rule.

The literature review could also be broader. I also think the literature review could be a little broader. The paper cites hierarchical bandits and uplift modeling, but because the authors emphasize that the environment is highly dynamic, I expected a deeper discussion of non-stationary bandits and adaptation methods. That would make the positioning of the paper stronger.

Minor Comments

• Some wording is repetitive, especially in the discussion of reward inheritance.

• Figure 2 would be clearer with one concrete end-to-end example.

References

Agrawal, A., Mohanty, R., Bhattacharjee, S., & Mittal, A. (2026). Hierarchical contextual uplift bandits for catalog personalization [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2601.14333